

PEC CH-2 Assignment Amplitude Modulation (AM) Communication

Q.1 What is modulation? why it is needed?

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Modulation :- modulation એ એક પ્રકાર છે જેમાં High-Frequency Carrier signal ના એક અથવા બે ગુણકો સાથે low-Frequency msg signal ના અવકાશીય ગુણો સંકલિત થાય છે.

Need of Modulation :-

1. Reduced Antenna Size :- Antenna ની minimum height $H = \frac{\lambda}{4}$ જેવી જરૂરી છે. જો direct 20KHz ની audio signal transmit કરીએ તો antenna ની height 3750 m થશે જે practically possible નથી. Modulation ની Frequency વધારીને antenna નું size ઘટાડી શકાય છે.

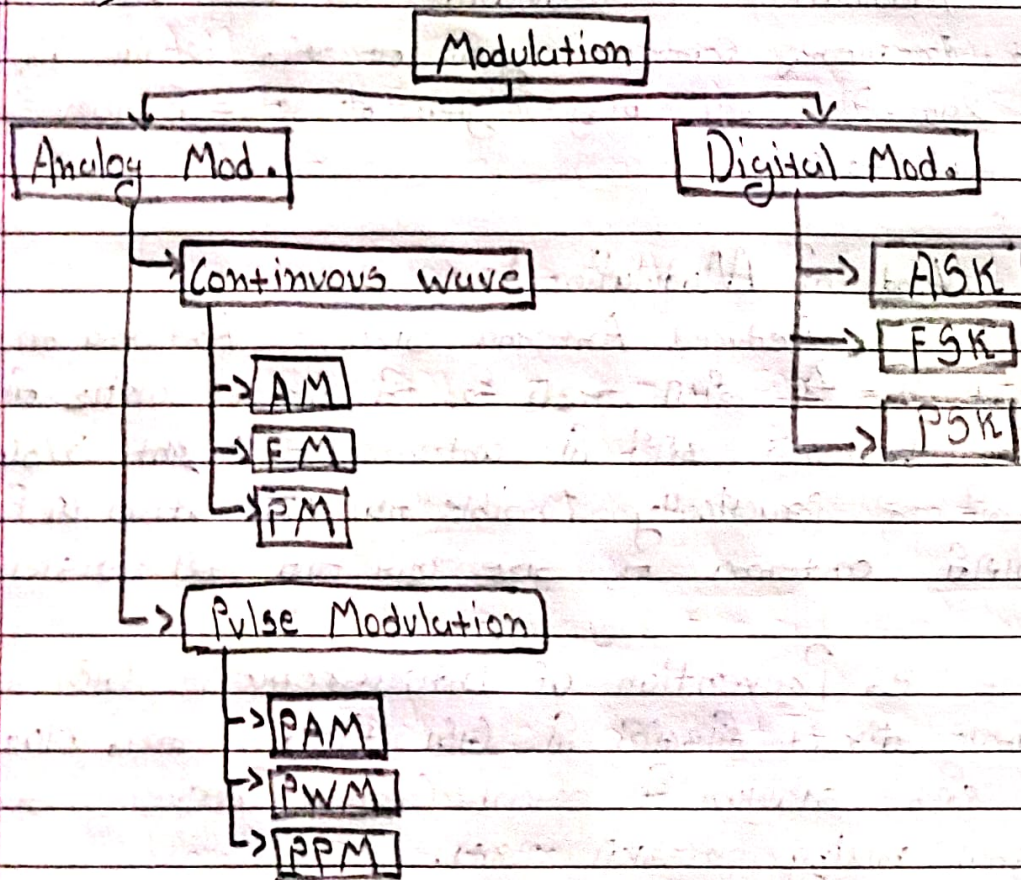
2. Prevention of Interference :- જો audio signals direct મોકલીએ તો જો સમયે જરૂર નથી, Modulation ની દરેક station ને અલગ Carrier Frequency મળે છે જેથી Signal mixing બચાવી શકાય.

3. Multiplexing :- એક જ Common channel પરથી બે અથવા વધુ જોડા સાથે મોકલી શકાય છે modulation કરીને.

4. Efficient Long Distance Transmission :- low frequency baseband signal લોકો દૂર મોકલવાની જરૂર નથી, high frequency Carrier સાથે તે easily long distance travel કરે છે.

5. Increased Noise Immunity :- Modulated signal is external noise and distortion of carrier signal is not affected.

Q.2 Classify modulation techniques.



• Analogy Modulation :- Continuous Wave Modulation and Pulse Modulation को समझाया गया है.

• Digital Modulation :- Digital Data Transmission में ASK, FSK, PSK और PCM का उपयोग.

Q.3 Draw and explain Amplitude Modulation and Derive Mathematical equation of AM signal.

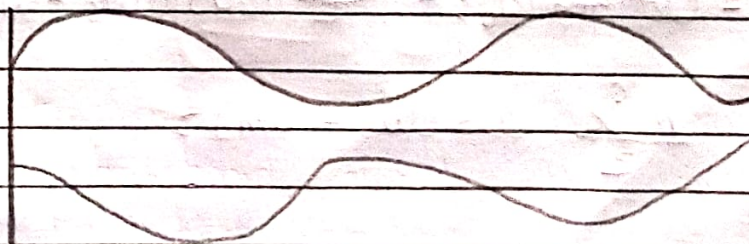
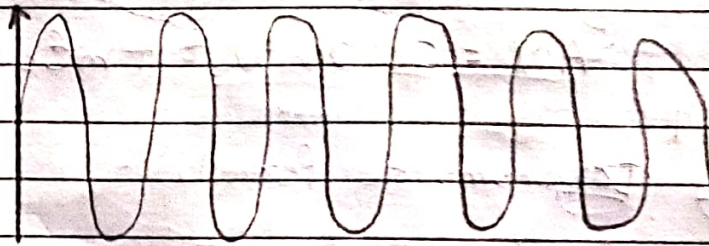
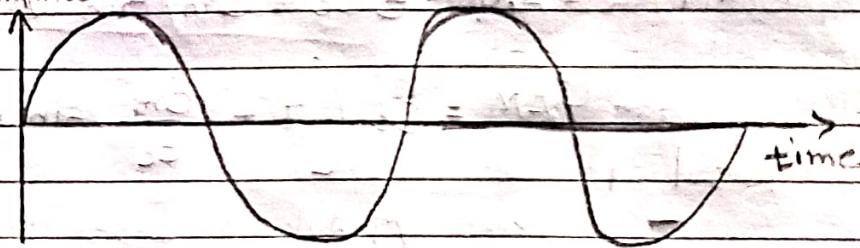
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Explanation :- Amplitude modulation is a high frequency carrier wave of Amplitude message signal of amplitude A_m change with ω_m and frequency ω_c and phase constant ϕ_c .

Waveforms :-

• Message Signal

Amplitude



• Mathematical Derivation :-

Example 3,

• Carrier wave: $E_c = E_c (\sin \omega_c t)$

• Modulating signal: $E_m = E_m \sin \omega_m t$

AM wave of instantaneous amplitude

$$A = E_c + E_m \sin \omega_m t \text{ V.A.}$$

AM signal of general equation:

• $E_{AM} = (E_c + E_m \sin \omega_m t) \sin \omega_c t$

• $E_{AM} = E_c \left(1 + \frac{E_m}{E_c} \sin \omega_m t \right) \sin \omega_c t$

- Taking Modulation Index $m = \frac{E_m}{E_c}$:

• $E_{AM} = E_c (1 + m \sin \omega_m t) \sin \omega_c t$

• $E_{AM} = E_c \sin \omega_c t + m E_c \sin \omega_m t \sin \omega_c t$

Using trigonometric formula :-

$$\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)] :$$

$$E_{AM} = E_c \sin \omega_c t + \frac{m E_c}{2} \cos(\omega_c - \omega_m)t -$$

$$\frac{m E_c}{2} \cos(\omega_c + \omega_m)t$$

• AM Components:-

- Carrier Component: $E_c \sin \omega_c t$

- Lower Sideband (LSB):
 $m E_c / 2 \cos(\omega_c - \omega_m) t$

- Upper Sideband (USB):
 $m E_c / 2 \cos(\omega_c + \omega_m) t$

4. Types of AM band spectrum like DSBFC, SSBSC, VSBSC and their application.

Parameter	DSBFC	DSBSC	SSBSC	VSB
Carrier Suppression	N.A.	Fully Suppressed	Fully Suppressed	Fully Suppressed
Sideband Suppression	N.A.	N.A.	One Sideband completely Suppressed	One Sideband Partially Suppressed
Bandwidth	$2f_m$	$2f_m$	f_m	$f_m < BW < 2f_m$
Transmission Efficiency	Minimum	Moderate	Maximum	Moderate
Complexity	Simple	Simple	Complex	Simpler than SSB

Q5. Derive Power relation in AM wave.

→ AM in total power is carrier power and Sidebands all power is sum of:

$$P_T = P_c + P_{USB}$$

in antenna resistance R is:

$$P_c = \frac{V_{rms}^2}{R} = \frac{(V_c / \sqrt{2})^2}{R} = \frac{V_c^2}{2R}$$

$$P_{USB} = P_{LSB} = \frac{(m_a V_c / 2\sqrt{2})^2}{R} = \frac{m_a^2 V_c^2}{8R}$$

Total power Equation:-

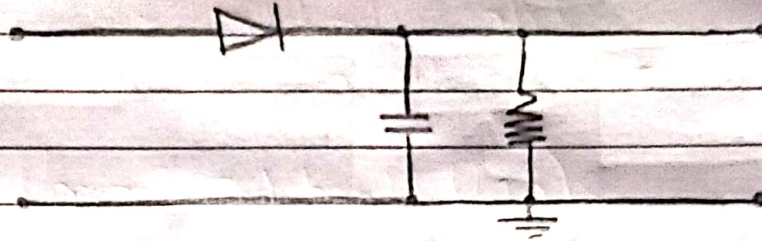
$$P_T = \frac{V_c^2}{2R} + \frac{m_a^2 V_c^2}{8R} + \frac{m_a^2 V_c^2}{8R}$$

$$P_T = \frac{V_c^2}{2R} + \frac{m_a^2 V_c^2}{4R}$$

$$= \frac{V_c^2}{2R} \left(1 + \frac{m_a^2}{2} \right)$$

$$P_T = P_c \left(1 + \frac{m_a^2}{2} \right)$$

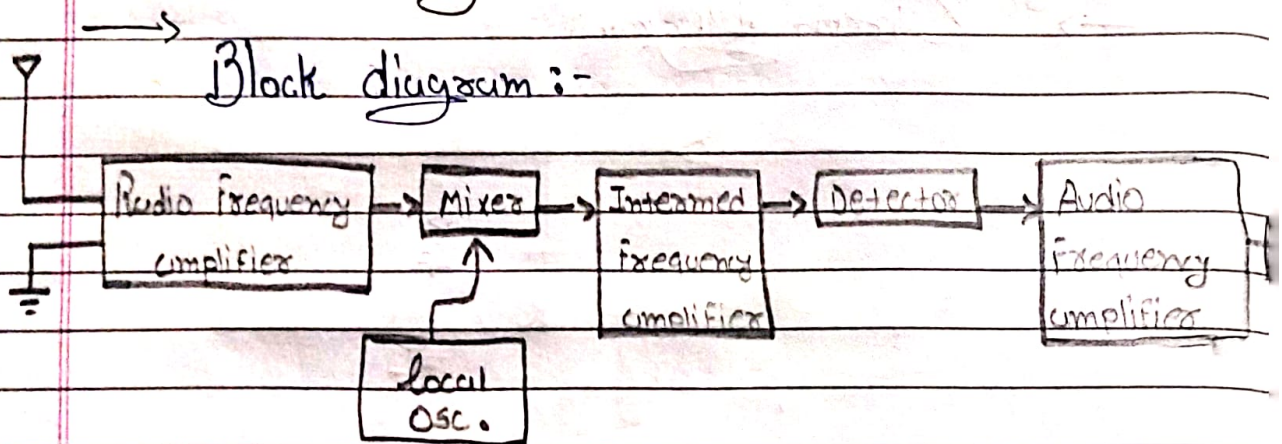
Q.6 Draw and Explain Envelope detector using diode.
→ Circuit Diagram:



Working Steps :-

1. Rectification :- Diode half-wave rectifier તરીકે કામ કરે છે અને યોગ્ય positive half cycle માં conduct થાય છે.
2. Charging :- જ્યારે input AM signal capacitor voltage કરતા વધે, ત્યારે diode forward bias થાય અને capacitor C ની Peak value તરફ Fast charge કરે છે.
3. Discharging :- Peak પછી diode reverse bias થાય અને capacitor resistor R through slowly discharge થાય છે.
4. Tracking :- આ Continuous charging and discharging cycle AM wave ની outer boundary ની track કરે છે.
5. Filtering :- RC low-pass filter high-frequency carrier ની bypass કરી original audio message signal output માં આપે છે.

Q.7 Explain Draw and explain block diagram of Superheterodyne receiver with Waveforms.



Working of each block:

1. Antenna :- Space and radio waves capture and electrical signal convert to.

2. RF Stage :- Desired signal (f_s) select to f_s , initial amplification and f_s and image frequency reject to f_s .

3. Mixer & local Oscillator :- local oscillator signal generate to f_s and mixer convert mixing and constant intermediate frequency convert to.

4. IF Amplifier :- fixed IF frequency is signal to f_s gain and sharp selectivity provide to f_s .

5. Detector :- IF signal aur main modulating audio signal ki demodulate krte hai.

6. Automatic Gain Control :- Input signal fluctuate krte hai uski speaker ki constant volume maintain krne ki iske gain adjust krte hai.

7. Audio Amp & Speaker :- Audio power boost krte hai aur speaker through electrical signal ki sound waves ki convert krte hai.

Q.8. Explain Characteristics of receiver.
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- Selectivity :- Receiver ki audio capacity aur iski help krke unwanted station signals aur ki desired frequency station ki select krte hai aur iski help krke reject krte hai aur iske IF amplifier stage ki depend krte hai.

- Sensitivity :- Receiver ki weak aur weak signal ki capture krne ki properly amplify krne ki ability. iski RF aur IF stages ki gain ki depend krte hai.

- Fidelity :- Transmitter aur iski modulated audio signals ki help krke frequency components ki help krke distortion output ki as-it-is reproduce krne ki receiver ki capacity.

• Image Frequency Rejection:- Mixer is a circuit unwanted image frequency is received at enter and also the ability of the RF amplifier stage of quality we depend on it.